

Unit 2 - Search Engine Optimization (SEO)

Digital Marketing (DI03016011) - GTU Diploma IT Sem 3

24% weightage | 8 hours

How to use these notes: each topic is written the way you can copy it in the exam. **3 marks** = definition + 2-3 points. **4 marks** = add example or table. **7 marks** = write everything, with diagram/table and a real example. Lines marked ★ show where the topic was asked in a GTU paper (only 2 papers exist: Winter 2025 and Summer 2026).

Questions already asked from this unit (do these first)

- S2026 Q2(a) [3M: need of SEO in digital marketing]
- BOTH papers: W2025 Q2(c) [7M: how search engines work + ranking factors] + S2026 Q2(b)OR [4M: algorithms and ranking]
- BOTH papers: W2025 Q2(b) [4M: white vs black hat with example] + S2026 Q3(b) [4M: explain black hat and white hat techniques]
- S2026 Q2(b) [4M: on-page vs off-page SEO]
- S2026 Q2(c) [7M: off-page optimisation for a newly launched website, with examples]
- S2026 Q3(b)OR [4M: strategies for building backlinks]
- W2025 Q2(b)OR [4M: how mobile-friendliness and page speed influence SEO ranking]
- W2025 Q2(c)OR [7M: steps to improve SEO ranking of a new blog organically]
- W2025 Q1(c) [7M: short SEO strategy for a local shoe-repair business, 3 on-page + 3 off-page]
- W2025 Q1(c)OR [7M: keyword strategy + Meta Description and Title Tag for 'Custom Cakes' page of 'Sweet Delights']
- S2026 Q2(a)OR [3M: role of social media in digital marketing]
- S2026 Q2(c)OR [7M: how social media can improve SEO rankings, with example]
- W2025 Q2(a) [3M: zero-click searches and their significance in SEO]

2.1 Overview and need of SEO

★ **S2026 Q2(a) [3M: need of SEO in digital marketing]**

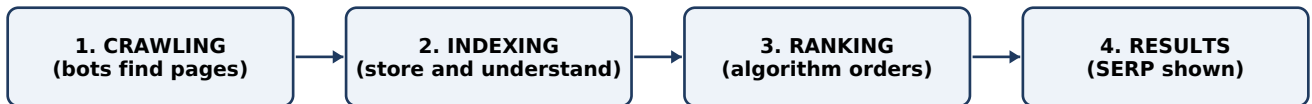
SEO (Search Engine Optimization) is the process of improving a website so that it ranks higher in the **free (organic)** results of search engines like Google.

Need / importance

- **More organic traffic** - most people click only the top results on page 1.
- **Cost-effective** - no payment per click, long-lasting results.
- **Trust and credibility** - users trust high-ranking sites.
- **Targeted visitors** - people searching for exactly your product.
- Better **user experience** (fast, mobile-friendly site) and higher conversions.
- Helps beat competitors and supports local business (near-me searches).

2.1 How search engines work + ranking factors

★ BOTH papers: W2025 Q2(c) [7M: how search engines work + ranking factors] + S2026 Q2(b)OR [4M: algorithms and ranking]



- **Crawling:** automated programs called **crawlers / spiders / bots** (Googlebot) travel from link to link and discover new or updated pages. A **sitemap.xml** helps them; **robots.txt** tells them what not to crawl.
- **Indexing:** the content of each page (text, images, titles, links) is analysed and saved in a huge database called the **index**.
- **Ranking:** when a user searches, the **algorithm** picks relevant pages from the index and orders them using hundreds of signals. Best results are shown first on the **SERP** (search engine results page).

Key factors that influence ranking

Factor	Why it matters
Content quality and relevance	Useful, original content that matches the user's search intent ranks higher
Keywords	Right keywords in title, headings and body show what the page is about
Backlinks	Links from trusted sites act as 'votes' of trust (authority)
Mobile-friendliness	Google uses mobile-first indexing
Page speed	Slow pages lose visitors and rank lower
HTTPS security	Secure sites are preferred
User experience	Low bounce rate, easy navigation, longer time on site
Freshness and structured data	Regularly updated pages; schema helps rich results
Location and personalisation	Results change by user's location and search history

Exam tip: 4 marks: write the 4-step flow with 1 line each and 4-5 factors. 7 marks: full answer with the table.

2.2 White hat and black hat SEO

★ BOTH papers: W2025 Q2(b) [4M: white vs black hat with example] + S2026 Q3(b) [4M: explain black hat and white hat techniques]

Point	White Hat SEO	Black Hat SEO
Meaning	Ethical techniques that follow search-engine guidelines	Unethical tricks that try to fool search engines for fast ranking
Focus	Real users and quality content	Only search-engine algorithm
Techniques	Quality content, proper keyword research, good titles and meta tags, mobile-friendly site, genuine backlinks, fast loading	Keyword stuffing, cloaking, hidden text, link farms / buying links, doorway pages, copied / spun content
Result	Slow but long-term, stable ranking	Quick gain but short-lived
Risk	Very low	Penalty or complete removal (ban) from search results
Example	A travel blog writes original guides and gets links from other travel sites	A site repeats 'cheap shoes cheap shoes cheap shoes' 100 times or hides white text on a white background

Grey hat SEO lies in between (not clearly breaking rules).

2.3 On-page, off-page and technical SEO

★ S2026 Q2(b) [4M: on-page vs off-page SEO]

Point	On-page SEO	Off-page SEO	Technical SEO
Meaning	Optimising things on the website itself	Activities outside the website to build authority	Optimising the site's back-end so it can be crawled and indexed
Main elements	Title tag, meta description, headings, keywords, quality content, URL, image alt text, internal links	Backlinks, guest posts, social sharing, brand mentions, directory listings, influencer outreach	Site speed, mobile-friendly, HTTPS, sitemap.xml, robots.txt, schema markup, fixing broken links
Control	Fully in your control	Partly (depends on others)	In your control
Goal	Relevance	Authority and trust	Crawlability and performance
Example	Adding 'custom cakes' in title tag	Getting a link from a food blog	Compressing images to load faster

2.3 Off-page optimisation for a new website

★ S2026 Q2(c) [7M: off-page optimisation for a newly launched website, with examples]

Off-page SEO = actions taken **outside** your website to raise its authority, trust and ranking. For a new website (no authority yet):

Activity	How to do it	Example
Quality backlinks	Earn links from relevant, trusted sites	A new fitness website gets a link from a health blog
Guest blogging	Write useful articles on other blogs with a link back	Guest post on a popular fitness blog
Social media promotion	Share content, build a following, drive traffic	Post articles on Instagram, LinkedIn, X, Facebook
Business directories / listings	Submit to directories and Google Business Profile with same NAP	Justdial, Google Maps
Influencer and PR outreach	Collaborate with influencers; press release for launch	Local news site covers the launch
Community engagement	Answer questions on Quora / Reddit / forums with helpful links	Quora answer with a link to your guide
Reviews and brand mentions	Ask customers for reviews; get brand talked about	Google reviews, testimonials
Video / content sharing	Upload videos, infographics, slides	YouTube video linking to your site

- Focus on **quality over quantity**; avoid link farms and paid links (black hat).
- Analyse competitors' backlinks with free tools such as **openlinkprofiler.org**.
- Measure with domain authority, number of referring domains and referral traffic.

2.3 Backlink building strategies

★ S2026 Q3(b)OR [4M: strategies for building backlinks]

A **backlink** is a link from another website to yours. Search engines treat quality backlinks as votes of trust. Strategies:

1. **Create link-worthy content:** original research, infographics, guides, free tools.
2. **Guest blogging:** write for related websites and include a link.
3. **Broken-link building:** find broken links on other sites and suggest your page as replacement.

4. **Outreach and influencers:** ask bloggers, journalists and partners to mention you.
5. **Directory and local listings** and social profiles.
6. **Competitor backlink analysis** to find where competitors get links (openlinkprofiler.org, Ahrefs).
7. **Press releases and PR** for news-worthy events.

Avoid buying links or link farms (black hat) - they can lead to penalties.

2.4 Mobile-friendliness and page speed

★ W2025 Q2(b)OR [4M: how mobile-friendliness and page speed influence SEO ranking]

- **Mobile-friendliness:** most searches are on phones and Google uses **mobile-first indexing** (the mobile version of the page is used for ranking). A non-responsive site with tiny text and hard-to-tap buttons gives a poor experience and ranks lower.
- **Page speed:** speed is a ranking signal and part of **Core Web Vitals** (loading, interactivity, visual stability). A slow site increases the **bounce rate**, lowers time on site and conversions, which tells Google the page is not useful.
- **How to improve:** responsive design, compress images, enable caching / CDN, minify CSS and JavaScript, use a good host. Test with Google PageSpeed Insights, GTmetrix.

Example: If an online shop takes 7 seconds to load on mobile, most visitors leave; after compressing images and using responsive design it loads in 2 seconds, so traffic and ranking improve.

2.4 Techniques to improve ranking: new blog

★ W2025 Q2(c)OR [7M: steps to improve SEO ranking of a new blog organically]

1. **Keyword research** - find topics people search for (Google Keyword Planner, Ubersuggest); prefer long-tail low-competition keywords.
2. **Create high-quality, original content** that answers the user's question; write regularly.
3. **On-page SEO** - keyword in title (under 60 chars), meta description (under 160), headings, URL, image alt text; short readable paragraphs.
4. **Internal linking** between related posts.
5. **Technical SEO** - mobile-friendly theme, fast loading, HTTPS, sitemap.xml; submit the blog in **Google Search Console**.
6. **Build backlinks** - guest posts, share in communities, ethical outreach, link-worthy content (infographics).
7. **Promote on social media** to get traffic and shares.
8. **Update old posts**, fix broken links, improve slow pages.
9. **Track** traffic, bounce rate and rankings in Analytics / Search Console and keep improving.

Use only organic (free, white-hat) methods - no paid ads, no buying links.

Applied SEO: local shoe repair business

★ W2025 Q1(c) [7M: short SEO strategy for a local shoe-repair business, 3 on-page + 3 off-page]

Goal: appear in the top results when people near the shop search 'shoe repair near me' and bring walk-ins and calls.

Step 1: Keyword research

Use Google Keyword Planner / Ubersuggest. Target local, long-tail keywords: *shoe repair near me*, *sole repair [city]*, *leather shoe polish and repair*, *bag zip repair*.

Step 2: On-page tactics (at least three)

1. **Title tag and meta description** with keyword + city, e.g. 'Shoe Repair in [City] | Quick Sole and Leather Repair'.
2. **Service pages and headings** (H1, H2): separate page for sole repair, stitching, polishing, with local keywords and prices.

3. **Image alt text and quality photos** of before / after repairs; fast, mobile-friendly pages.
4. **LocalBusiness schema** with name, address, phone (NAP) and opening hours; a clear call button.

Step 3: Off-page tactics (at least three)

1. **Google Business Profile:** claim and complete it (address, timings, photos). Most important for local SEO.
2. **Customer reviews:** ask happy customers to review on Google and reply to each.
3. **Local directories / citations** (Justdial, Sulekha, Google Maps) with the **same NAP** everywhere.
4. **Local backlinks:** links from local blogs, colleges, shoe shops and community pages.
5. **Social media** pages with before / after posts and local hashtags.

Step 4: Measure

Track calls, direction requests, rankings for 'shoe repair near me', and website visits in Google Search Console and Analytics.

Applied SEO: keywords, title tag and meta description (bakery)

★ W2025 Q1(c)OR [7M: keyword strategy + Meta Description and Title Tag for 'Custom Cakes' page of 'Sweet Delights']

Keyword strategy

1. **Seed keywords:** custom cakes, designer cakes, birthday cakes, wedding cakes.
2. **Long-tail and local keywords** (easier to rank, buying intent): 'custom birthday cake in [city]', 'eggless custom cake near me', 'photo cake delivery', 'theme cake for kids'.
3. **Search intent:** informational (cake design ideas), commercial ('best custom cake shop'), transactional ('order custom cake online'). The Custom Cakes page targets **transactional** keywords.
4. **Tools:** Google Keyword Planner, Ubersuggest; check search volume, competition and what competitors use.
5. **Placement:** main keyword in title tag, H1, first 100 words, URL (/custom-cakes), image alt text; related keywords in sub-headings. No keyword stuffing.
6. Use separate blog pages for informational keywords and link them to the Custom Cakes page.

Title tag (max 60 characters)

Custom Cakes Near You | Sweet Delights Bakery (45 characters)

Meta description (max 160 characters)

Order fresh, eggless and designer custom cakes for birthdays and weddings from Sweet Delights. Choose flavour, size and design online. Order today! (147 characters)

Both contain the main keyword, are unique, and end with a call to action. Replace 'Near You' with the city name for local ranking.

2.5 Role of social media in SEO

★ S2026 Q2(a)OR [3M: role of social media in digital marketing]

★ S2026 Q2(c)OR [7M: how social media can improve SEO rankings, with example]

- **Brand awareness:** reaches millions of users quickly.
- **Engagement:** two-way contact through likes, comments, shares and messages.
- **Traffic and leads:** drives visitors to the website, lead forms.
- **Customer service and feedback** in real time.
- **Targeted advertising** using age, interest, location.
- **Insights:** analytics show what content works.

Google says social media signals (likes, shares) are **not a direct ranking factor**, but social media **helps SEO indirectly**:

1. **More traffic and visibility:** shared content gets more visitors, which can lead to more engagement and links.
2. **Backlinks:** content that goes viral is picked up by bloggers and websites who link to it - strong ranking signal.
3. **Brand awareness and brand searches:** more people search for your brand name, which signals popularity and authority.
4. **Faster content discovery:** new posts shared on social media are found and crawled sooner.
5. **Social profiles rank** in search results and take more space on the first page for your brand name.
6. **Local SEO:** reviews and location tags support local ranking.
7. **Content distribution and audience insight:** see which topics people like and write SEO content on them.

Example: A bakery posts a Reel of a new cake and shares the blog link. The post goes viral, a food blogger links to the blog, many people search 'Sweet Delights cakes', and the website ranks higher for related keywords.



Zero-click searches

★ W2025 Q2(a) [3M: zero-click searches and their significance in SEO]

Zero-click search: a search where the user gets the answer **directly on the search results page** (featured snippet, knowledge panel, weather, calculator, map pack) and **does not click** any website.

Significance in SEO

- Organic **click-through rate and website traffic fall**, even if the page ranks well.
- Brand gets **visibility and authority** if its content is shown in the featured snippet (position zero).
- SEO must now optimise for **snippets**: short direct answers, headings as questions, FAQ and schema markup.
- Success is measured by **impressions and brand searches**, not only clicks.

Last-minute revision sheet - Unit 2

- SEO = improve organic ranking. Needs: free traffic, trust, targeted users.
- Search engine: Crawl → Index → Rank → Results.
- White hat = ethical; black hat = keyword stuffing, cloaking, link farms (penalty risk).
- On-page (title, meta, content), Off-page (backlinks, social), Technical (speed, sitemap, HTTPS).
- Title tag max 60 characters, meta description max 160.
- Social media helps SEO indirectly (traffic, backlinks, brand searches).